

Introduction to Electrodynamics Syllabus

2025 2nd semester

Title	Introduction to Electrodynamics		Code		Grades	3	Classroom	
			Course	Elective	Hours	3		
Instructor	Name	Class	Office	Office hour	Contact	e-mail		
	김현벽		Chang 5306			kimhyunbyuk@ksa.hs.kr		

1. Outline

Course Objective
After completing this course, students should be able to • understand the mathematical tools for Maxwell's equations. • understand the physical meanings and applications of Maxwell's equations.
Course Description
▪ 3-dimensional vector calculus : Vector and Scalar functions for several variables will be introduced. Several kinds of spatial differentiation will be treated. Several integral theorems also will be discussed. ▪ Static cases of Maxwell's equation : Field and potential distributions for the static cases will be discussed. Especially, the effects of the presence of material will be treated. ▪ Dynamic cases of Maxwell's equation : Radiation and Electromagnetic wave phenomena will be discussed. Special Relativistic features also will be checked.

2. Core Target Competencies

Critical Thinking	Analytical and Creative Thinking	Integrity
Basic knowledge is dealt in the class. But the main point is not to deliver the established facts but to train the thinking method. Students are heavily encouraged to sharpen their critical arguments and in-depth discussion skills. Moreover, having a firm big picture of the subject is one of the ultimate goals of the course.		

3. Text

- A. Textbook : Introduction to Electrodynamics **5th** (Griffiths, Cambridge)
 B. Reference : The Feynman Lectures on Physics (Feynman *et al.*, Basic Books)

4. Class Method

- A. Lecture, Discussion & Debate

5. Evaluation (%)

Midterm	Final	Quiz	Homework☆	Attendance & Attitude*	Total
30	30	10	20	10	100

☆ Penalty will be given to the overdue homework. : -50% after due date

※ Penalty : - 0.3 point per late & - 1.0 point per absence without notice

※※ 공결 및 병결 등의 사항은 학업성적관리규정을 따름

6. Grade (절대평가)

$A+ \geq 80\%$, $80\% > A0 \geq 75\%$, $75\% > A- \geq 70\%$,
 $70\% > B+ \geq 65\%$, $65\% > B0 \geq 60\%$, $60\% > B- \geq 55\%$,
 $55\% > C+ \geq 50\%$, $50\% > C0 \geq 45\%$, $45\% > C- \geq 40\%$,
 $40\% > D+ \geq 35\%$, $35\% > D0 \geq 30\%$, $30\% > D- \geq 25\%$, $25\% > F$

7. Schedule

Week	Subject	Homework
1	Ch1. Vector Analysis	Handout
	Analytical Thinking, Integrity	
2	Ch2. Electrostatics	Handout
	Critical Thinking, Analytical and Creative Thinking, Integrity	
3	Ch3. Potentials	Handout
	Critical Thinking, Analytical and Creative Thinking, Integrity	
4	Ch4. Electric Fields in Matter	Handout
	Critical Thinking, Analytical and Creative Thinking, Integrity	
5	Ch5. Magnetostatics	Handout
	Critical Thinking, Analytical and Creative Thinking, Integrity	
6	Ch6. Magnetic Fields in Matter	Handout
	Critical Thinking, Analytical and Creative Thinking, Integrity	
7	Ch7. Electrodynamics	Handout
	Critical Thinking, Analytical and Creative Thinking, Integrity	
8	연휴	
9	Midterm Exam	
10	Ch7. Electrodynamics	Handout
	Critical Thinking, Analytical and Creative Thinking, Integrity	
11	Ch8. Conservation Laws	Handout
	Critical Thinking, Analytical and Creative Thinking, Integrity	
12	Ch9. Electromagnetic Waves	Handout
	Critical Thinking, Analytical and Creative Thinking, Integrity	
13	Ch10. Potentials and Fields	Handout
	Critical Thinking, Analytical and Creative Thinking, Integrity	
14	Ch11. Radiation	Handout
	Critical Thinking, Analytical and Creative Thinking, Integrity	
15	Ch11. Radiation	Handout
	Critical Thinking, Analytical and Creative Thinking, Integrity	
16	Ch12. Electrodynamics and Relativity	보강 주간
	Critical Thinking, Analytical and Creative Thinking, Integrity	
17	Final Exam	

*** Sections will be selective for some chapters.

*** Schedules can be changed according to the circumstances.